

Loops and Stopping

Project: Transparent Book Recommender

1. Loops and Stopping

Teach the learning cycle

Primary target: Trace variable values before and after each loop iteration

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/4, ai-2/5, python-bridge/6. Reuse a familiar

case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Trace Tables

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

Look for: Trace variable values before and after each loop iteration

Reference example, not the only acceptable wording: 2, 3 and 4; the stop value is excluded. The body runs three times.

Decompose a Goal

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a function definition from a function call

Reference example, not the only acceptable wording: Name show_remaining; parameters capacity and used. Arguments 15 and 6 match capacity = 15 and used = 6. Expected printed result 9.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Trace `count=0; while count<4: count=count+2`. List values tested by the condition and final count. Would replacing `<4` with `!=5` terminate?

Concept to explain

Useful repetition needs a clear update and stopping rule.

Term · Meaning

Term: loop

Meaning: A structure that repeats instructions according to a count or condition.

Term: update

Meaning: Change a stored value or state according to a specified operation or new evidence.

Term: stopping rule

Meaning: A specified condition that ends a repeated process.

Term: infinite loop

Meaning: A loop that never reaches its stopping condition under the assumed execution model.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach for loops with finite integer ranges before adding more loop forms. Keep range values, body executions and output events separate. Reference: <https://docs.python.org/3/tutorial/controlflow.html#the-range-function>. The examples use a positive default step and do not require lists or while loops.

Readiness answer

It reads the old 2, adds 3 and stores 5. Assignment updates a value; it is not an algebraic equality.

Python 3 editor and loops.py

Counters or numbered paper cards

Blank trace table with turn total before and total after columns

End goal for the pictorial case

By the end, I can trace a while-loop and recognise when its update cannot satisfy the stopping test.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

2. Repeat while count < 4: show a card; count = count + 1.

3. The loop runs 4 times and stops when count becomes 4.

4. If count never changes, the stopping condition never becomes false.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Start n=0; while n<3, add 1 and record n.

List records and stopping condition.

Exact explanation to model

1,2,3. The next comparison is false. The update makes progress toward termination.

Explain the added worked example

Imagine adding numbered counters: first 1, then 2, then 3. A loop can repeat the same addition with a different number each time. This week you will predict each change before running a short Python script. Tracing repeated calculations prepares you to inspect the data processing used in larger AI applications.

In for turn in range(1, 4), Python supplies 1, 2 and 3 in order. The start is included and the stop is excluded. The indented body runs once for each supplied value. The colon and four-space indentation mark that body. This lesson uses integer range values with the default positive step of 1.

The output is 6. Before the first iteration, total is 0. Each assignment reads the previous total and adds the current turn. After turn 3, the range is exhausted, so execution continues after the loop. The table records state; it does not create extra instructions.

Key terms

Term · Meaning

Term: Loop

Meaning: A block of instructions that can run repeatedly

Term: Iteration

Meaning: One execution of a loop body

Term: Accumulator

Meaning: A variable that keeps a running result

Term: Initialization

Meaning: Giving a variable its starting value

Term: Trace table

Meaning: A record of variable values as instructions execute

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

range(1, 4): 1, 2, 3 · 3

range(1, 4): range(1, 2)

1, 2, 3: 1

3: 1

range(1, 4): range(1, 1)

1, 2, 3: No values

3: 0

range(1, 4): range(4)

1, 2, 3: 0, 1, 2, 3

3: 4

With one argument, range starts at zero. range(4) supplies four values but none is 4. If the worked script uses range(1, 1), the body never runs and the initialized total remains 0. Do not rely on a variable first assigned inside a loop: a zero-iteration run might never assign it.

Replace range(1, 4) with range(1, 3) in the worked script. Predict the supplied values and final

total. What stop value is needed to include 3?

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Loops and Stopping

A fictional counter starts at 12 and decreases while above a boundary.

The facts we will use

Python: `count=12; while count>3: count=count-4; print after the loop.`

The condition is checked before each subtraction.

Loops and Stopping

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial check

$12 > 3$: true; subtract 4 to get 8.



2. Next check

$8 > 3$: true; subtract 4 to get 4.



3. Final update

$4 > 3$: true; subtract 4 to get 0.



4. Stop

$0 > 3$ is false; report 0 after three updates.

CHANGE ONE THING

Replace $\text{count} > 3$ with $\text{count} \neq 3$.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Initial check

What happens and why: $12 > 3$: true; subtract 4 to get 8.

Step: 2. Next check

What happens and why: $8 > 3$: true; subtract 4 to get 4.

Step: 3. Final update

What happens and why: $4 > 3$: true; subtract 4 to get 0.

Step: 4. Stop

What happens and why: $0 > 3$ is false; report 0 after three updates.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The condition does not promise the counter ends exactly at 3. Subtractions can cross the boundary. The count strictly decreases, so the below-or-equal stopping region is eventually reached from this start. A condition requiring exact equality can be unsafe when repeated updates skip that value.

What this example does not establish

Do not run an intentionally unbounded loop in class; trace a finite prefix to inspect the defect safely.

Change one thing in the pictured case

Change: Replace $\text{count} > 3$ with $\text{count} \neq 3$.

Prediction: From 12, the loop never reaches 3 and does not terminate in this toy model.

Reason: 12, 8, 4, 0, -4, ... are multiples of 4; none equals 3.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Start $n=0$; while $n < 3$, add 1 and record n .

Write the question to explain: List records and stopping condition.

Reveal after predictions: 1, 2, 3. The next comparison is false. The update makes progress toward termination.

Keep for an independent attempt: Start at count 0. While count is less than 3, add 1 and display count. List the displays and the stopping value.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

`range(1, 3)` supplies 1 and 2, giving total 3. Stop 4 includes 3. Exit: the reset bug agrees for one iteration because there are no earlier additions to lose. `range(1, 4)` should give 6; the faulty version gives 3.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve assignment using the prerequisite.

Stage: 7 to 20 minutes

Teacher action: Model range with numbered cards and complete W1.

Stage: 20 to 33 minutes

Teacher action: Trace each row of the worked addition and predict output.

Stage: 33 to 45 minutes

Teacher action: Complete W2 and compare predicted and observed values.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve included start and excluded stop.

Stage: 8 to 20 minutes

Teacher action: Trace the reset bug, complete W3 and discuss the guided response.

Stage: 20 to 35 minutes

Teacher action: Collect independent W4 code and zero one and several iteration traces.

Stage: 35 to 45 minutes

Teacher action: Use W5 and the exit to examine assumptions and revealing tests.

Reduce help deliberately

Completion task: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each iteration. Predict the single output, then run and compare.

Teacher check: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final output 9. Keep the observed result distinct from the prediction.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Loops and trace tables. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: List the supplied values before tracing; range(1, 4) excludes 4.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why is report 0 correct for the original loop, and why is equality-only stopping risky here?

Reasoned teacher answer

The original stops as soon as count is not above 3. Exact equality fails because the update sequence skips 3.

Changed-case reasoning: From 12, the loop never reaches 3 and does not terminate in this toy model.
12,8,4,0,-4,... are multiples of 4; none equals 3.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Walk a robot around a four-card loop. Record the update each time; remove the update to observe the failure; restore it and retest.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The stop value is included: List the supplied values before tracing; range(1, 4) excludes 4.

Initialization belongs inside every iteration: Resetting the accumulator loses earlier work.

An unindented print repeats with the body: Indentation determines which statements repeat.

Zero iterations mean the entire program does nothing: Initialization and statements after the loop still execute.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

1,2,3. The next comparison is false. The update makes progress toward termination.

Independent question

Start at count 0. While count is less than 3, add 1 and display count. List the displays and the stopping value.

Expected independent answer

Displays are 1, 2 and 3. The next condition check finds 3 is not less than 3, so the loop stops.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A less-than-3 loop can never display 3.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

The condition tests 0,2,4; the body runs twice and ends at 4. With $!=5$, even counts never equal 5, so it does not terminate under unbounded arithmetic. A stopping rule should match the update and include resource bounds in an actual application.

Explanations for the additional workbook examples

W1 Explain the range

List the values supplied by `range(2, 5)` and explain why 5 is absent. How many times does its body run?

2, 3 and 4; the stop value is excluded. The body runs three times.

W2 Trace before running

Copy the worked script. Change the range to `range(2, 5)`. Record turn, total before and total after each iteration. Predict the single output, then run and compare.

Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final output 9. Keep the observed result distinct from the prediction.

W3 Repair the reset

For the faulty script with `range(1, 4)`, record total after each addition. Explain the wrong final output and remove only the faulty reset.

Faulty after-addition values are 1, 2, 3 because each starts from zero. Remove the indented `total = 0`. Correct after-addition values are 1, 3, 6; final output 6.

W4 Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use `rounds = 3`, an initialized total and for turn in `range(rounds)`. Print the final total once after the loop. Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

Initialize `total = 5` before the loop; add 2 inside it. For three rounds, turn values 0, 1, 2 give totals 7, 9, 11. Final outputs for 0, 1, 3 rounds are 5, 7, 11. Print is unindented.

W5 Check a contract limitation

In your counter script, `range(-2)` performs no iterations. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

No. The written contract requires a nonnegative integer. The script assumes this and does not reject negative values. Empty range behavior does not validate the input; report the limitation or add an explicit check.

Assess independent transfer

Selected case: Start count at 1. While `count < 8`, add 3. Trace the condition and values. What happens if the condition becomes `count != 8`?

Selected case answer

The condition is true at 1,4,7, then false at 10. Final count is 10. With `!=8` the sequence never reaches 8: all values are 1 modulo 3, so this loop does not terminate.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Transparent Book Recommender

Loop through every candidate book, update its score, then stop at the end of the list.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 6: Debugging Strategies. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use counters and let the learner move a pointer through one instruction at a time. Offer a blank trace grid and permit oral explanation. Extension: put print inside the correct loop and explain why zero rounds

produces no output while an outside print still does.

W4 must preserve the initial 5, update by 2, trace three iterations and print once. Require outputs 5, 7, 11 for zero one and three rounds. After support, retry with an initial 4 and additions of 3 for zero one and two rounds: outputs 4, 7, 10. Record assistance before judging independence.

Week 7 stores several values in a list and distinguishes an index from the value at that position.

Plan the delayed return

Suggested return points in this level: ai-2/6, ai-2/8, ai-2/11. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Use progress and stopping arguments for any repeated recommender process; test inputs that require zero, one and several updates.

7. Prepare, teach and assess

Teaching target: Trace variable values before and after each loop iteration

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each iteration. Predict the single output, then run and compare.

Reasoned practice answer: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final output 9. Keep the observed result distinct from the prediction.

Check every learner before discussion: Use the worked case for Loops and trace tables. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: List the supplied values before tracing; $\text{range}(1, 4)$ excludes 4.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace $\text{count} > 3$ with $\text{count} \neq 3$. Require a prediction and a reason before comparing. Expected result: From 12, the loop never reaches 3 and does not terminate in this toy model. Reason: 12, 8, 4, 0, -4, ... are multiples of 4; none equals 3.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner’s explanation and evidence.